

Limits (Part IV)

Limits at Removable Discontinuities (cont.)

Factoring the Difference/Sum of Two Cubes

Factoring the Sum or Difference of Two Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Find the values of the limits algebraically

$$\bullet \textcircled{1} \lim_{k \rightarrow \frac{3}{2}} \frac{8k^3 - 27}{2k - 3} = \lim_{k \rightarrow \frac{3}{2}} \frac{(2k)^3 - 3^3}{2k - 3} = \lim_{k \rightarrow \frac{3}{2}} \left\{ \frac{(2k-3)[(2k)^2 + 2k(3) + 3^2]}{2k-3} \right\}$$

$$a=2k, b=3 = \lim_{k \rightarrow \frac{3}{2}} (4k^2 + 6k + 9)$$

$$\begin{array}{r} 4k^2 + 6k + 9 \\ 2k-3 \overline{) 8k^3 - 27} \\ \underline{8k^3 - 12k^2} \\ 12k^2 - 27 \\ \underline{12k^2 - 18k} \\ 18k - 27 \\ \underline{18k - 27} \\ 0 \end{array}$$

$$= 4\left(\frac{3}{2}\right)^2 + 6\left(\frac{3}{2}\right) + 9$$

$$= 9 + 9 + 9$$

$$= \boxed{27}$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$\textcircled{2} \lim_{s \rightarrow -1} \frac{s+1}{s^3+1} = \lim_{s \rightarrow -1} \frac{s+1}{s^3+1^3} = \lim_{s \rightarrow -1} \left[\frac{s+1}{(s+1)(s^2-s(1)+1^2)} \right]$$

$$a=s, b=1$$

$$\begin{array}{r} s^2 - s + 1 \\ s+1 \overline{) s^3 + 1} \\ \underline{s^3 + s^2} \\ -s^2 + 1 \\ \underline{-s^2 - s} \\ s + 1 \\ \underline{s + 1} \\ 0 \end{array}$$

$$= \lim_{s \rightarrow -1} \frac{1}{s^2 - s + 1}$$

$$= \frac{1}{(-1)^2 - (-1) + 1}$$

$$= \frac{1}{1 + 1 + 1}$$

$$= \boxed{\frac{1}{3}}$$

$$\begin{array}{c} -1 \mid 1 \quad 0 \quad 0 \quad 1 \\ \quad \quad -1 \quad 1 \quad -1 \\ \quad \quad \downarrow \quad \downarrow \quad \downarrow \\ \quad \quad s^2 - s + 1 \end{array}$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$\textcircled{3} \lim_{r \rightarrow -\frac{4}{5}} \frac{5r+4}{125r^3+64} = \lim_{r \rightarrow -\frac{4}{5}} \frac{5r+4}{(5r)^3+4^3} = \lim_{r \rightarrow -\frac{4}{5}} \left\{ \frac{5r+4}{(5r+4)[(5r)^2-5r(4)+4^2]} \right\}$$

$$a=5r, b=4$$

$$\begin{array}{r} 5r+4 \overline{) 125r^3+64} \\ \underline{125r^3+100r^2} \\ -100r^2+64 \\ \underline{-100r^2-80r} \\ 80r+64 \\ \underline{80r+64} \\ 0 \end{array}$$

$$= \lim_{r \rightarrow -\frac{4}{5}} \frac{1}{25r^2-20r+16}$$

$$= \frac{1}{25(-\frac{4}{5})^2 - 20(-\frac{4}{5}) + 16}$$

$$= \frac{1}{16 + 16 + 16}$$

$$= \boxed{\frac{1}{48}}$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$\textcircled{4} \lim_{m \rightarrow 6} \frac{m^3-216}{m-6} = \lim_{m \rightarrow 6} \frac{m^3-6^3}{m-6} = \lim_{m \rightarrow 6} \left[\frac{(m-6)(m^2+m(6)+6^2)}{m-6} \right]$$

$$a=m, b=6$$

$$\begin{array}{r} m-6 \overline{) m^3-216} \\ \underline{m^3-6m^2} \\ 6m^2-216 \\ \underline{6m^2-36m} \\ 36m-216 \\ \underline{36m-216} \\ 0 \end{array}$$

$$= \lim_{m \rightarrow 6} (m^2 + 6m + 36)$$

$$= 6^2 + 6(6) + 36$$

$$= 36 + 36 + 36$$

$$= \boxed{108}$$

$$\begin{array}{r} m-6 \overline{) m^3-216} \\ 6 \overline{) 1 \ 0 \ 0 \ -216} \\ \underline{6 \ 36 \ 216} \\ 1 \ 6 \ 36 \ 0 \\ \downarrow \downarrow \downarrow \\ m^2 + 6m + 36 \end{array}$$

Using Trigonometric Identities

Ratio Identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Reciprocal Identities

$$\sin \theta = \frac{1}{\csc \theta} \quad \csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{1}{\sec \theta} \quad \sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{1}{\cot \theta} \quad \cot \theta = \frac{1}{\tan \theta}$$

Double Angle Identities

$$\sin(2A) = 2 \sin A \cos A$$

$$\cos(2A) = \cos^2 A - \sin^2 A$$

$$\cos(2A) = 1 - 2 \sin^2 A$$

$$\cos(2A) = 2 \cos^2 A - 1$$

Opposite Angle Identities

$$\sin(-\theta) = -\sin \theta \quad \cos(-\theta) = \cos \theta \quad \tan(-\theta) = -\tan \theta$$

$$\csc(-\theta) = -\csc \theta \quad \sec(-\theta) = \sec \theta \quad \cot(-\theta) = -\cot \theta$$

Find the values of the limits algebraically.

$$\bullet \textcircled{5} \lim_{x \rightarrow 0} \frac{\tan x}{\sin x} = \lim_{x \rightarrow 0} \left(\frac{\frac{\sin x}{\cos x}}{\sin x} \right) = \lim_{x \rightarrow 0} \left(\frac{\sin x}{\cos x} \div \sin x \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x}{\cos x \sin x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{\cos x}$$

$$= \lim_{x \rightarrow 0} \sec x$$

$$= \sec(0)$$

$$= \boxed{1}$$

$$\bullet \textcircled{6} \lim_{\beta \rightarrow \frac{\pi}{2}} \frac{\cos \beta}{\cot \beta} = \lim_{\beta \rightarrow \frac{\pi}{2}} \left(\frac{\cos \beta}{\frac{\cos \beta}{\sin \beta}} \right) = \lim_{\beta \rightarrow \frac{\pi}{2}} \left(\cos \beta \cdot \frac{\sin \beta}{\cos \beta} \right)$$

$$= \lim_{\beta \rightarrow \frac{\pi}{2}} \left(\frac{\cos \beta}{1} \cdot \frac{\sin \beta}{\cos \beta} \right)$$

$$= \lim_{\beta \rightarrow \frac{\pi}{2}} \left(\frac{\cos \beta \sin \beta}{\cos \beta} \right)$$

$$= \lim_{\beta \rightarrow \frac{\pi}{2}} \sin \beta$$

$$= \sin \frac{\pi}{2}$$

$$= \boxed{1}$$

$$\bullet \textcircled{7} \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\sin \theta - 1}{\cos^2 \theta} = \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\sin \theta - 1}{1^2 - \sin^2 \theta} = \lim_{\theta \rightarrow \frac{\pi}{2}} \left[\frac{\sin \theta - 1}{(1 - \sin \theta)(1 + \sin \theta)} \right]$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$= \lim_{\theta \rightarrow \frac{\pi}{2}} \left[\frac{-1(-\sin \theta + 1)}{(1 - \sin \theta)(1 + \sin \theta)} \right]$$

$$= \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{-1}{1 + \sin \theta}$$

$$= \frac{-1}{1 + \sin \frac{\pi}{2}}$$

$$= \frac{-1}{1 + 1}$$

$$= \boxed{-\frac{1}{2}}$$

$$\bullet \textcircled{8} \lim_{A \rightarrow 0} \frac{\sin^2 A}{1 - \cos A} = \lim_{A \rightarrow 0} \frac{1 - \cos^2 A}{1 - \cos A} = \lim_{A \rightarrow 0} \left[\frac{(1 - \cos A)(1 + \cos A)}{1 - \cos A} \right]$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$= \lim_{A \rightarrow 0} (1 + \cos A)$$

$$= 1 + \cos(0)$$

$$= 1 + 1$$

$$= \boxed{2}$$

$$\bullet \textcircled{9} \lim_{x \rightarrow 0} \frac{\sin x}{\sin(2x)} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{2 \sin x \cos x} \right) = \lim_{x \rightarrow 0} \frac{1}{2 \cos x} = \frac{1}{2 \cos(0)}$$

$$\sin(2x) = 2 \sin x \cos x$$

$$= \frac{1}{2(1)}$$

$$= \boxed{\frac{1}{2}}$$

★ There are three identities; you must choose the right one.

$$\textcircled{10} \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos(2x)}{\cos x - \sin x} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos^2 x - \sin^2 x}{\cos x - \sin x} = \lim_{x \rightarrow \frac{\pi}{4}} \left[\frac{(\cos x - \sin x)(\cos x + \sin x)}{\cos x - \sin x} \right]$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} (\cos x + \sin x)$$

$$= \cos \frac{\pi}{4} + \sin \frac{\pi}{4}$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}$$

$$= \frac{2}{\sqrt{2}}$$

$$= \frac{2 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}}$$

$$= \frac{2\sqrt{2}}{\sqrt{4}}$$

$$= \frac{2\sqrt{2}}{2}$$

$$= \boxed{\sqrt{2}}$$

$$\bullet \textcircled{11} \lim_{P \rightarrow -\frac{\pi}{2}} \left[\frac{\cos P \sin P}{\cos(-P)} \right] = \lim_{P \rightarrow -\frac{\pi}{2}} \left[\frac{\cos P \sin P}{\cos P} \right] = \lim_{P \rightarrow -\frac{\pi}{2}} \sin P = \sin\left(-\frac{\pi}{2}\right) = \boxed{-1}$$

$$\bullet \textcircled{12} \lim_{\alpha \rightarrow \pi} \left(\frac{\sec \alpha \sin(-\alpha)}{\sin \alpha} \right) = \lim_{\alpha \rightarrow \pi} \left[\frac{\sec \alpha (-\sin \alpha)}{\sin \alpha} \right] = \lim_{\alpha \rightarrow \pi} \left(\frac{-\sec \alpha \sin \alpha}{\sin \alpha} \right)$$

$$= \lim_{\alpha \rightarrow \pi} (-\sec \alpha)$$

$$= -\sec \pi$$

$$= -(-1)$$

$$= \boxed{1}$$

The Limit Laws

If both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, and "c" is a constant, then

$$1) \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$2) \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$3) \lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$4) \lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \quad \lim_{x \rightarrow a} g(x) \neq 0$$

$$5) \lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x)$$

$$6) \lim_{x \rightarrow a} c = c$$

$$7) \lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) \text{ if "f" is continuous where its input is } \lim_{x \rightarrow a} g(x).$$

8) If "f" is continuous at $x=a$, then $\lim_{x \rightarrow a} f(x) = f(a)$.
 *Remember, this is the definition of continuity at $x=a$.

Special Limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$$

More Challenging Problems Requiring the Limit Laws

Find the value of each limit. Write the number of the limit law used on each step.

$$\bullet \textcircled{13} \lim_{x \rightarrow 0} \frac{3x^2 - 9x}{\sin x} = \lim_{x \rightarrow 0} \left[\frac{x(3x-9)}{\sin x} \right] = \lim_{x \rightarrow 0} \left[\frac{x}{\sin x} \cdot \frac{(3x-9)}{1} \right]$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\stackrel{3}{=} \left(\lim_{x \rightarrow 0} \frac{x}{\sin x} \right) \left(\lim_{x \rightarrow 0} (3x-9) \right)$$

$$\stackrel{8}{=} \left(\lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}} \right) (3(0)-9)$$

$$\stackrel{4}{=} \frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} (0-9)$$

$$\stackrel{6}{=} \frac{1}{1} (-9)$$

$$= \boxed{-9}$$

$$\textcircled{14} \lim_{w \rightarrow 0} \frac{\cos^2 w - 1}{w} = \lim_{w \rightarrow 0} \frac{\cos^2 w - 1^2}{w} = \lim_{w \rightarrow 0} \left[\frac{(\cos w - 1)(\cos w + 1)}{w} \right]$$

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$$

$$= \lim_{w \rightarrow 0} \left[\frac{\cos w - 1}{w} \cdot \frac{\cos w + 1}{1} \right]$$

$$\stackrel{3}{=} \left(\lim_{w \rightarrow 0} \frac{\cos w - 1}{w} \right) \left(\lim_{w \rightarrow 0} (\cos w + 1) \right)$$

$$\stackrel{8}{=} (0) (\cos(0) + 1)$$

$$= (0) (1 + 1)$$

$$= (0) (2)$$

$$= \boxed{0}$$

$$\begin{aligned}
 \textcircled{15} \lim_{B \rightarrow 0} \frac{-2B}{\sin^2 B + 3 \sin B} &= \lim_{B \rightarrow 0} \left[\frac{-2B}{\sin B (\sin B + 3)} \right] \\
 &= \lim_{B \rightarrow 0} \left[\frac{B}{\sin B} \cdot \frac{-2}{\sin B + 3} \right] \\
 &\stackrel{3}{=} \lim_{B \rightarrow 0} \frac{B}{\sin B} \lim_{B \rightarrow 0} \frac{-2}{\sin B + 3} \\
 &= \left(\lim_{B \rightarrow 0} \frac{1}{\frac{\sin B}{B}} \right) \left(\lim_{B \rightarrow 0} \frac{-2}{\sin B + 3} \right) \\
 &\stackrel{4 \& 8}{=} \left(\frac{\lim_{B \rightarrow 0} 1}{\lim_{B \rightarrow 0} \frac{\sin B}{B}} \right) \left(\frac{-2}{\sin(0) + 3} \right) \\
 &\stackrel{6}{=} \left(\frac{1}{1} \right) \left(\frac{-2}{0 + 3} \right) \\
 &= (1) \left(-\frac{2}{3} \right) \\
 &= \boxed{-\frac{2}{3}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{16} \lim_{x \rightarrow 0} \left[\frac{(1 - \cos x)^2}{3x} \right] &= \lim_{x \rightarrow 0} \left[\frac{(1 - \cos x)(1 - \cos x)}{3x} \right] = \lim_{x \rightarrow 0} \left[\left(\frac{1 - \cos x}{3} \right) \left(\frac{1 - \cos x}{x} \right) \right] \\
 &\stackrel{3}{=} \left(\lim_{x \rightarrow 0} \frac{1 - \cos x}{3} \right) \left(\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \right) \\
 &\stackrel{8}{=} \left(\frac{1 - \cos(0)}{3} \right) (0) \\
 &= \left(\frac{1 - 1}{3} \right) (0) \\
 &= \left(\frac{0}{3} \right) (0) \\
 &= (0)(0) \\
 &= \boxed{0}
 \end{aligned}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\bullet \textcircled{17} \lim_{x \rightarrow 0} \left(\frac{3x^3}{\sin^3 x} \right) = \lim_{x \rightarrow 0} \left[3 \cdot \frac{x^3}{(\sin x)^3} \right] \stackrel{5}{=} 3 \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right)^3$$

$$\stackrel{7}{=} 3 \left(\lim_{x \rightarrow 0} \frac{x}{\sin x} \right)^3$$

$$= 3 \left(\lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}} \right)^3$$

$$\stackrel{4}{=} 3 \left(\frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} \right)^3$$

$$\stackrel{6}{=} 3 \left(\frac{1}{1} \right)^3$$

$$= 3(1)^3$$

$$= 3(1)$$

$$= \boxed{3}$$

$$\bullet \textcircled{18} \lim_{B \rightarrow 0} \frac{\sin B}{B + \sin B} = \lim_{B \rightarrow 0} \left(\frac{\frac{\sin B}{B}}{\frac{B + \sin B}{B}} \right) = \lim_{B \rightarrow 0} \left(\frac{\frac{\sin B}{B}}{\frac{B}{B} + \frac{\sin B}{B}} \right)$$

$$= \lim_{B \rightarrow 0} \left(\frac{\frac{\sin B}{B}}{1 + \frac{\sin B}{B}} \right)$$

$$\stackrel{4}{=} \frac{\lim_{B \rightarrow 0} \frac{\sin B}{B}}{\lim_{B \rightarrow 0} \left(1 + \frac{\sin B}{B} \right)}$$

$$\stackrel{1}{=} \frac{1}{\lim_{B \rightarrow 0} 1 + \lim_{B \rightarrow 0} \frac{\sin B}{B}}$$

$$\stackrel{6}{=} \frac{1}{1 + 1}$$

$$= \boxed{\frac{1}{2}}$$

$$\bullet \textcircled{19} \lim_{x \rightarrow 0} \frac{\cos x - 1}{\sin x} = \lim_{x \rightarrow 0} \left(\frac{\frac{\cos x - 1}{x}}{\frac{\sin x}{x}} \right) \stackrel{4}{=} \frac{\lim_{x \rightarrow 0} \frac{\cos x - 1}{x}}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} = \frac{0}{1} = \boxed{0}$$

For $\lim_{x \rightarrow 0} \frac{\sin(kx)}{kx}$, $k = \text{constant}$, let $\theta = kx$, then $\lim_{x \rightarrow 0} \frac{\sin(kx)}{kx} = \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

$$\bullet \textcircled{20} \lim_{x \rightarrow 0} \frac{\sin(4x)}{x} = \lim_{x \rightarrow 0} \left[\frac{4 \cdot \sin(4x)}{4 \cdot x} \right] = \lim_{x \rightarrow 0} \left[\frac{4 \cdot \sin(4x)}{4x} \right]$$

$$\stackrel{5}{=} 4 \lim_{x \rightarrow 0} \left[\frac{\sin(4x)}{4x} \right]$$

Let $A = 4x$

$$= 4 \lim_{A \rightarrow 0} \left(\frac{\sin A}{A} \right)$$

$$= 4(1)$$

$$= \boxed{4}$$

$$\bullet \textcircled{21} \lim_{t \rightarrow 0} \frac{\sin(\frac{1}{2}t)}{t} = \lim_{t \rightarrow 0} \left[\frac{\sin(\frac{1}{2}t)}{t} \right] = \lim_{t \rightarrow 0} \left[\frac{\frac{1}{2} \cdot \sin(\frac{1}{2}t)}{\frac{1}{2} \cdot t} \right]$$

$$= \lim_{t \rightarrow 0} \left[\frac{\frac{1}{2} \cdot \sin(\frac{1}{2}t)}{\frac{1}{2}t} \right]$$

$$\stackrel{5}{=} \frac{1}{2} \lim_{t \rightarrow 0} \left[\frac{\sin(\frac{1}{2}t)}{\frac{1}{2}t} \right]$$

$$\text{Let } A = \frac{1}{2}t \quad = \frac{1}{2} \lim_{A \rightarrow 0} \frac{\sin A}{A}$$

$$= \frac{1}{2}(1)$$

$$= \boxed{\frac{1}{2}}$$

$$\begin{aligned}
 \bullet \text{ (2d) } \lim_{x \rightarrow 0} \frac{x\sqrt{x+9}}{\tan(6x)} &= \lim_{x \rightarrow 0} \frac{x\sqrt{x+9}}{\frac{\sin(6x)}{\cos(6x)}} = \lim_{x \rightarrow 0} \left[x\sqrt{x+9} \div \frac{\sin(6x)}{\cos(6x)} \right] \\
 &= \lim_{x \rightarrow 0} \left[\frac{x\sqrt{x+9}}{1} \cdot \frac{\cos(6x)}{\sin(6x)} \right] \\
 &= \lim_{x \rightarrow 0} \left[\frac{x\sqrt{x+9} \cos(6x)}{\sin(6x)} \right] \\
 &= \lim_{x \rightarrow 0} \left[\frac{x}{\sin(6x)} \cdot \sqrt{x+9} \cos(6x) \right] \\
 &\stackrel{3}{=} \lim_{x \rightarrow 0} \frac{x}{\sin(6x)} \lim_{x \rightarrow 0} [\sqrt{x+9} \cos(6x)] \\
 &\stackrel{8}{=} \lim_{x \rightarrow 0} \left[\frac{1}{\frac{6 \sin(6x)}{6 \cdot x}} \right] [\sqrt{0+9} \cos(6(0))] \\
 &\stackrel{4 \& 5}{=} \frac{\lim_{x \rightarrow 0} 1}{6 \lim_{x \rightarrow 0} \left[\frac{\sin(6x)}{6x} \right]} [\sqrt{9} \cos(0)] \\
 &\stackrel{6}{=} \frac{1}{6 \lim_{A \rightarrow 0} \frac{\sin A}{A}} [3(1)]
 \end{aligned}$$

Let $A = 6x$

$$= \frac{1}{6(1)} [3]$$

$$= \frac{1}{6} (3)$$

$$= \frac{3}{6}$$

$$= \boxed{\frac{1}{2}}$$

$$\bullet \textcircled{23} \lim_{\theta \rightarrow 0} \frac{\tan(5\theta)}{\theta} = \lim_{\theta \rightarrow 0} \left[\frac{\frac{\sin(5\theta)}{\cos(5\theta)}}{\theta} \right] = \lim_{\theta \rightarrow 0} \left[\frac{\sin(5\theta)}{\cos(5\theta)} \div \theta \right]$$

$$= \lim_{\theta \rightarrow 0} \left[\frac{\sin(5\theta)}{\cos(5\theta)} \div \frac{\theta}{1} \right]$$

$$= \lim_{\theta \rightarrow 0} \left[\frac{\sin(5\theta)}{\cos(5\theta)} \cdot \frac{1}{\theta} \right]$$

$$= \lim_{\theta \rightarrow 0} \left[\frac{\sin(5\theta)}{\theta \cos(5\theta)} \right]$$

$$= \lim_{\theta \rightarrow 0} \left[\frac{\sin(5\theta)}{\theta} \cdot \frac{1}{\cos(5\theta)} \right]$$

$$\stackrel{3}{=} \lim_{\theta \rightarrow 0} \frac{\sin(5\theta)}{\theta} \lim_{\theta \rightarrow 0} \sec(5\theta)$$

$$\stackrel{8}{=} \lim_{\theta \rightarrow 0} \left[\frac{5 \cdot \sin(5\theta)}{5 \cdot \theta} \right] \sec(5(0))$$

$$= \lim_{\theta \rightarrow 0} \left[\frac{5 \cdot \sin(5\theta)}{5\theta} \right] (\sec(0))$$

$$\stackrel{5}{=} 5 \lim_{\theta \rightarrow 0} \left[\frac{\sin(5\theta)}{5\theta} \right] (1)$$

Let $A = 5\theta$

$$= 5 \lim_{A \rightarrow 0} \frac{\sin A}{A}$$

$$= 5(1)$$

$$= \boxed{5}$$